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Chemical Thermal Performance of Nonmetallic Polymers, PEEK, PVDF, PPS and SMPs in Extreme Downhole Conditions

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Abstract

This paper investigates the potential application of advanced nonmetallic polymers—PEEK (Polyetheretherketone), PVDF (Polyvinylidene Fluoride), and PPS (Polyphenylene Sulfide)—alongside Shape Memory Polymers (SMPs) in downhole completion tools. The focus is on assessing their chemical properties, mechanical strength, and thermal stability as alternatives to conventional metallic components in high-pressure, high-temperature (HPHT) environments, where challenges such as corrosion, erosion, and mechanical failures commonly arise.

A comparative analysis was conducted to evaluate the intrinsic properties of PEEK, PVDF, PPS, and SMPs against conventional metallic materials in downhole completion systems. Mechanical strength and thermal stability were quantified using standard industry metrics such as tensile modulus and decomposition temperature. The study examines their suitability for replacing metal-based components in key applications such as packers, sealing elements, and sand screens. The analysis emphasizes the polymers' resistance to chemical degradation by hydrocarbons, CO₂, H₂S, and aggressive acids, as well as their performance under thermal cycling and dynamic movement. SMPs were specifically considered for their ability to change properties in response to temperature stimuli, highlighting their potential in specialized downhole applications.

The findings suggest that PEEK is highly suitable for packers and sealing elements, due to its superior thermal stability (up to 334°C / 633°F) and strong chemical resistance to hydrocarbons and sour gases, reducing risks of corrosion and material degradation. PVDF, known for its exceptional resistance to aggressive acids and bases, shows promise for use in sand screens, particularly where chemical exposure is severe, despite its lower temperature limit (170°C / 338°F). PPS, with a high melting point (285°C / 545°F) and robust chemical resistance, is ideal for sealing applications in high-acidity environments. Additionally, Shape Memory Polymers (SMPs) exhibit adaptive behavior in response to temperature changes, making them effective for sand management and zonal isolation in downhole completions.

This paper presents a detailed evaluation of nonmetallic polymers as viable replacements for traditional metallic components in downhole completion tools. The enhanced chemical and thermal properties of PEEK, PVDF, and PPS offer significant potential for improving the performance and lifespan of composite sand screens, addressing common issues like corrosion and erosion. The study's insights pave the way for

broader applications of these materials, highlighting opportunities for future innovations in inflow control devices (ICDs) and inflow control valves (ICVs), expanding the operational capabilities of nonmetallic technologies in HPHT environments.

Introduction

Background

As oil and gas developments extend into deeper, more geochemically hostile reservoirs, downhole completion tools are increasingly subjected to environments characterized by high pressure, high temperature (HPHT), and chemically aggressive fluids. These completion systems—which include packers, inflow control devices (ICDs), isolation valves, sealing elements, and sand control screens—are required to maintain long-term structural and sealing integrity under conditions that can exceed 20,000 psi and 200°C. In these conditions, conventional metallic components are vulnerable to stress corrosion cracking, scaling, erosion, and chemical degradation, particularly in the presence of hydrogen sulfide (H₂S), carbon dioxide (CO₂), and high total dissolved solids (TDS) ([Almuhammadi et al., 2023](#)).

The failure of these components can lead to costly workovers, production deferment, and environmental risks, prompting a paradigm shift in material selection strategies. Emerging evidence points to the inadequacy of traditional metallic alloys, especially in ultra-HPHT or high-H₂S fields, where even corrosion-resistant alloys (CRAs) may fail due to localized attack mechanisms ([Almuhammadi et al., 2023](#)).

Challenges with Metallic Components

While metallic alloys like Inconel and stainless steel have historically served as the materials of choice for downhole tools, they suffer from fundamental material limitations when exposed to dynamic HPHT environments. Problems such as hydrogen-induced cracking (HIC), sulfide stress cracking (SSC), and thermal fatigue persist across metallic systems, even when high-grade CRAs are employed ([Asthana et al., 2023](#)). In addition, metallic systems are often heavy, rigid, and prone to rapid corrosion in acidic or sour gas conditions, adding weight to the completion string and complexity to the deployment.

Further, the mechanical behavior of metallic components under multi-axial loading and long-term thermal cycling can lead to premature seal failure, warping, or creep—especially in dynamic sealing applications. In many cases, seal stacks or metallic wear rings experience degradation well before the designed life of the tool is reached, particularly in open hole completions where tool conformability is essential ([Asthana et al., 2023](#)).

Motivation for Exploring Nonmetallic Polymers

The industry's response to these challenges has been a progressive move toward nonmetallic materials, particularly high-performance polymers, fluoropolymers, and thermally responsive materials such as shape memory polymers (SMPs). These materials offer advantages in terms of corrosion resistance, lower weight, and thermal and mechanical adaptability. In particular, polymers such as Polyetheretherketone (PEEK), Polyvinylidene Fluoride (PVDF), and Polyphenylene Sulfide (PPS) have demonstrated promising performance in sour and thermal applications, including seals, packers, and insulation systems ([Al Munif et al., 2025](#); [Asthana et al., 2023](#); [Almuhammadi et al., 2023](#)).

Shape memory polymers, meanwhile, introduce a new dimension of adaptability in well completions by enabling temperature-triggered expansion to conform to open hole geometries. When subjected to activation temperatures in the range of 90–150°C, SMPs can expand to fill voids, providing both mechanical support and sand control, while achieving filtration efficiencies on par with traditional gravel packs.

Motivated by this technological evolution, the present study investigates the performance of PEEK, PVDF, PPS, and SMPs in the context of downhole completion tools. Emphasis is placed on their chemical

compatibility, thermal stability, and mechanical suitability for HPHT environments and their comparative advantages over metallic counterparts.

Materials Overview

Polyetheretherketone (PEEK)

PEEK is a semi-crystalline thermoplastic widely used in downhole environments for its exceptional balance of strength, chemical resistance, and thermal stability. Its tensile strength typically ranges from 90 to 100 MPa, with a Young's modulus of approximately 3.6 GPa. PEEK's high melting point ($\sim 343^{\circ}\text{C}$) and continuous use temperature of up to 250°C allow it to perform reliably in high-temperature wells, while its low coefficient of thermal expansion (CTE) enhances dimensional stability in fluctuating thermal environments (Al Munif et al., 2025).

Advanced formulations of PEEK, including reinforced PolyArylEtherKetones, have shown capability to endure up to 288°C and 30,000 psi, with over 90% retention of tensile strength after 1,000 hours in high- H_2S sour gas environments (Chaplin et al., 2014). These properties position PEEK as a leading candidate for backup rings, seal carriers, and electrical insulators in HPHT applications.

Polyvinylidene Fluoride (PVDF)

PVDF is a fluoropolymer known for its excellent resistance to acids, bases, and hydrocarbons, particularly in sour service. While it has a lower mechanical strength compared to PEEK—tensile strength typically around 40–55 MPa—it maintains structural integrity up to approximately $130\text{--}150^{\circ}\text{C}$ depending on formulation and exposure duration. In testing under static exposure to H_2S - and CO_2 -rich fluids for 180 days, PVDF-lined carbon steel pipes showed no observable liner collapse or degradation at 130°C (Craster et al., 2020).

PVDF is increasingly used in cable insulation systems, gas barriers over EPDM insulation, and as a liner material in GRE pipelines and flexible composites for applications requiring acid resistance. However, its thermal limits and lower resistance to mechanical creep restrict its use in high-load structural applications compared to PEEK or PPS.

Polyphenylene Sulfide (PPS)

PPS is a high-performance thermoplastic with a high melting point ($\sim 285^{\circ}\text{C}$) and excellent resistance to oxidizing and reducing agents. It demonstrates chemical inertness to a wide range of substances including H_2S , amines, and salt brines. Although it has slightly lower mechanical strength than PEEK, PPS retains its dimensional integrity under compressive and tensile stresses in chemically reactive wells.

PPS has also been applied in hybrid seal systems and barrier materials in downhole tools, particularly in energized seal stacks for tubing/annulus barriers. PPS's durability in high- CO_2 , high-pressure applications has been validated through sour service simulations and long-term autoclave exposure tests.

Shape Memory Polymers (SMPs)

SMPs are a class of thermally responsive materials engineered to undergo pre-programmed shape changes when exposed to specific activation stimuli, typically temperature. In open hole sand control applications, SMP screens are deployed in a compacted state and thermally activated at the target depth using surface-pumped fluids heated to 150°C , after which they expand to conform to the borehole geometry (Wang and Osunjaye, 2016).

Once expanded, SMP structures demonstrate a post-deployment permeability of over 30 Darcy and offer filtration performance equivalent or superior to conventional open hole gravel packs. The material's porosity and ability to self-deploy without mechanical actuation make it an attractive solution for long-lateral and deviated wells where traditional completion equipment may be impractical or cost-prohibitive.

Comparative Evaluation Framework

Material selection for downhole completion components in HPHT environments could be guided by four primary performance criteria: thermal stability, mechanical strength, chemical resistance, and application-specific behavior such as sealing integrity or shape adaptability. Each of these criteria plays a distinct role in determining a polymer's viability for deployment under extreme temperature, pressure, and fluid exposure.

Thermal Performance

Thermal endurance is typically characterized by parameters such as melting point, glass transition temperature (T_g), and continuous use temperature. PEEK demonstrates exceptional thermal stability, with a melting point near 343°C and use temperatures exceeding 250°C. Reinforced grades of PEEK have been validated for operation up to 288°C while maintaining over 90% of their tensile strength after 1,000 hours in sour gas conditions (Chaplin et al., 2014). PPS offers similarly high thermal tolerance with a melting point near 285°C, although its T_g is lower, around 90°C. In contrast, PVDF's upper temperature limit is approximately 150°C, after which it undergoes rapid mechanical degradation (Craster et al., 2020). SMPs operate differently, with performance defined by their T_g or activation temperature, typically in the 90–150°C range, which determines their deployment behavior rather than their structural longevity.

Thermal cycling performance is also critical. Materials with high creep resistance and low coefficients of thermal expansion (CTE) are better suited for dynamic downhole environments. PEEK's low CTE ($\sim 46 \times 10^{-6}/^{\circ}\text{C}$) and high modulus help it maintain dimensional stability in thermal shocks (Al Munif, et al., 2023), while PVDF, with a higher CTE ($\sim 130 \times 10^{-6}/^{\circ}\text{C}$), is more prone to deformation under cyclic loads. PPS provides a favorable compromise with a moderate CTE ($\sim 60\text{--}70 \times 10^{-6}/^{\circ}\text{C}$) and improved structural rigidity under temperature fluctuation (Asthana et al., 2023).

Mechanical Properties

In high-stress sealing and load-bearing applications, mechanical strength and stiffness are key differentiators. PEEK exhibits tensile strengths in the range of 90–100 MPa and a Young's modulus of ~ 3.6 GPa, outperforming most other engineering polymers (Badeghaish et al., 2019). It also resists deformation under pressure and maintains shape over time, which is crucial for seal integrity in packers and completion valves. PPS offers slightly lower strength but maintains stiffness under thermal and chemical stress, making it suitable for secondary sealing and interface layers (Almuhammadi et al., 2023).

PVDF, while chemically robust, is mechanically softer, with tensile strengths around 40–55 MPa. It is more suitable for passive roles such as linings or insulation rather than dynamic sealing or load transfer (Craster et al., 2020). SMPs do not conform to typical mechanical benchmarks; their utility lies in controlled expansion and porosity rather than high strength. However, once expanded, SMP structures maintain enough integrity to serve as sand screens with permeability exceeding 30 Darcy (Wang and Osunjaye, 2016).

Chemical Compatibility

HPHT wells frequently expose downhole tools to aggressive chemistries, including high concentrations of CO_2 , H_2S , organic acids, brines, and hydrocarbons. PEEK is known for its broad-spectrum chemical resistance, with outstanding stability in sour environments and no degradation after exposure to high-partial-pressure CO_2 and H_2S (Chaplin, et al., 2014). PPS performs similarly, resisting acids, bases, and sulfidic media without significant mechanical compromise (Almuhammadi et al., 2023).

PVDF is optimized for acidic environments and has shown no collapse or delamination after 180-day exposure to sour gas at 130°C in lined pipe studies (Craster et al., 2020). However, it is not suitable for environments exceeding this thermal range. SMPs are less chemically inert compared to these thermoplastics, and while their short-term exposure behavior is acceptable, long-term chemical degradation studies remain limited in scope. Their primary qualification lies in successful deployment and retrieval rather than corrosion performance (Wang and Osunjaye, 2016).

Functional Behavior and Deployment Performance

Beyond bulk material properties, each polymer class brings application-specific behavior critical to tool function. PEEK's rigidity and high load capacity make it suitable for mechanically anchored sealing components and electrical insulators in ESPs and packers. PPS adds value in hybrid sealing stacks and chemical barrier interfaces, where thermal and chemical endurance is prioritized over structural rigidity (Asthana et al., 2023). PVDF's niche lies in non-load-bearing corrosion barriers, such as cable jacketing and inner liners in composite tubulars. It also supports chemical isolation in acid stimulation and water injector lines, where structural stress is minimal. SMPs enable a fundamentally different completion approach —offering temperature-triggered, conformable expansion for rigless zonal isolation and open hole sand control. The ability to adapt to wellbore irregularities without mechanical actuation presents a significant deployment advantage in extended laterals and shale formations (Wang and Osunjaye, 2016).

Table 1—Summary of Key Properties of Nonmetallic Polymers for Downhole Completion Tools

Property	PEEK	PVDF	PPS	SMPs
Melting Point (°C)	~343	~177	~285	Varies by formulation
Glass Transition Temp. (°C)	~143	~-35 to -40	~90	~90–120 (activation)
Max Continuous Use Temp. (°C)	250–288	130–150	200–240	90–150 (activation)
Tensile Strength (MPa)	90–100	40–55	70–100	Moderate (post-expansion)
Chemical Resistance	Excellent (H ₂ S, CO ₂ , hydrocarbons)	Excellent (acids, brines)	Excellent (acids, sour gas)	Moderate (short-term exposure)
CTE (×10 ⁻⁶ /°C)	~46	~130	~60–70	Moderate to high
Load Support	High	Low–Moderate	Moderate	Low (adaptive structure)
Key Applications	Seals, packers, insulators	Liners, cables, coatings	Barrier elements, hybrid seals	Sand screens, isolation tools
Notable Advantage	Thermal and chemical resilience	Acid resistance	Chemical durability in sour gas	Temperature-triggered expansion

Results and Discussion

PEEK: High Thermal Stability and Chemical Resistance

Among the materials evaluated, PEEK consistently demonstrates the most balanced profile for high-demand downhole applications. Its mechanical integrity under compressive and tensile stress, combined with outstanding resistance to sour fluids, supports its widespread use in high-pressure seal carriers, backup rings, and ESP insulation components. PEEK's glass transition temperature (~143°C) and melting point (~343°C) enable it to function in sustained service temperatures exceeding 250°C. Reinforced PEEK variants maintain structural strength at temperatures approaching 288°C and pressures up to 30,000 psi, with minimal loss of mechanical properties after 1,000 hours in sour gas autoclave testing (Chaplin et al., 2014).

Its low coefficient of thermal expansion (~46 × 10⁻⁶/°C) minimizes dimensional drift during thermal cycling, a crucial feature for dynamic sealing elements in packers and intervention-critical zones. Additionally, its resistance to hydrocarbons, acids, H₂S, and CO₂ across a broad pH range makes it an ideal candidate for chemical barrier roles in modular completion strings (Al Munif et al., 2025).

PVDF: Acid/Base Resistance with Thermal Limitations

PVDF demonstrates exceptional resistance to acidic environments, including HCl, HF, and high-TDS brines, making it well-suited for corrosion protection layers, cable insulation, and coated internal surfaces

of sand screens and tubulars. However, its performance envelope is thermally constrained. Under static conditions, PVDF maintains mechanical integrity up to 130–150°C, but creep deformation and reduced modulus limit its use in dynamic or high-load sealing configurations (Craster et al., 2020).

PVDF's chemical resistance and ease of processing make it particularly attractive for acid stimulation service, liner systems, and flow control tubing in corrosive but thermally moderate wells. It is not suitable for mechanically stressed components such as primary seals or pressure retention layers. Instead, it is best deployed in secondary containment roles or as a composite interface material in non-metallic or hybrid tubular systems.

PPS: High Melting Point and Chemical Robustness

PPS offers a strong alternative to PEEK in applications requiring high thermal tolerance with enhanced chemical inertness but lower mechanical loading. With a melting point of ~285°C and broad resistance to oxidizing and reducing agents, PPS performs well in high-H₂S and high-CO₂ gas wells, particularly where stress cracking and scale interaction are concerns (Almuhammadi et al., 2023). Its tensile strength and modulus are lower than PEEK but sufficient for use in sealing stack layers, chemical isolation components, and non-metallic packer elements.

PPS's intermediate coefficient of thermal expansion ($\sim 60\text{--}70 \times 10^{-6}/^\circ\text{C}$) and low moisture uptake also enhance its dimensional stability. While it is not typically used as a primary load-bearing material, it serves effectively in barrier rings, insulating sleeves, and composite seal assemblies, particularly where chemical aggression is expected but mechanical demand is moderate.

SMPs: Temperature-Responsive Behavior and Adaptive Sand Control

SMPs present a novel functional mechanism not found in conventional polymers. Their ability to undergo pre-programmed expansion upon reaching a defined activation temperature (typically 90–150°C) allows for deployment in open hole sand control and rigless isolation tools. Once exposed to activation fluid at depth, the SMP expands to match borehole geometry, creating mechanically anchored, porous structures with permeability exceeding 30 Darcy (Wang and Osunjaye, 2016).

This behavior eliminates the need for mechanical actuators, enabling simpler, lower-cost installations in long laterals, shale formations, or zones with access constraints. While SMPs are not intended for high-load applications, their thermal-driven adaptability makes them particularly useful in zonal isolation devices, adaptive screens, and temporary seals in expandable completion systems. Post-deployment mechanical strength is adequate for withstanding reservoir pressure and providing flow control without additional gravel packing.

Comparison with Metallic Materials

Compared to traditional metallic materials, all four polymers offer distinct advantages in corrosion resistance, chemical inertness, and design flexibility. PEEK and PPS, in particular, match or exceed metallic alloys in sour gas tolerance, while avoiding the weight, cost, and embrittlement issues commonly observed in CRAs. SMPs and PVDF, though not structurally equivalent to metals, offer deployment efficiencies and protection against failure modes such as acid corrosion, thermal creep, and scale adhesion. However, metallic components remain superior in extreme pressure retention and cyclic fatigue performance. Therefore, hybrid approaches—where nonmetallics are used strategically in seal zones, liners, or adaptive segments—represent the most practical transition toward broader nonmetallic adoption in HPHT completions.

Applications and Implications

The findings presented in this review support the targeted integration of nonmetallic polymers into downhole completion architectures, where traditional metallic systems face limitations due to corrosion, thermal

instability, and mechanical degradation. Each polymer demonstrates specific advantages when matched to application environments and tool requirements. Their implementation offers new opportunities in tool design, lifecycle reliability, and rigless deployment methods, particularly for HPHT wells and chemically aggressive reservoirs.

Packers and Zonal Isolation Devices

In packers, both static and dynamic sealing elements must endure radial and axial mechanical loads, elevated temperatures, and aggressive fluid chemistries. PEEK is a well-established choice for backup rings, structural inserts, and insulating components in both permanent and retrievable packers. Its combination of high tensile strength, dimensional stability, and chemical resistance makes it ideal for sealing stack carriers and mechanically loaded pressure containment rings in high-temperature packers (Al Munif et al., 2025; Chaplin et al., 2014).

PPS is increasingly adopted as a nonmetallic barrier material in multi-material sealing assemblies, especially in chemically aggressive completions where resistance to H₂S and amine-laden fluids is required. Its moderate stiffness and high thermal limit allow it to function as a secondary sealing component or interface element in composite packer systems (Asthana et al., 2023).

SMPs offer unique advantages in rigless zonal isolation, enabling expansion and conformance without mechanical actuation. SMP-based packers have been successfully deployed in field trials using thermal activation to expand and anchor across open hole sections, offering a low-intervention solution for multizone isolation in extended-reach wells (Wang and Osunjaye, 2016).

Sand Control Systems and Screens

SMPs are increasingly being applied in open hole sand control, replacing conventional gravel pack systems where deck space, lateral length, or complex well geometries make mechanical deployment impractical. Upon activation, the SMP expands and conforms to the borehole, maintaining over 30 Darcy permeability while providing mechanical anchoring and filtration without fluid loss or pack slumping. This allows the tool to perform both as a screen and a mechanical support, especially in shale zones, long horizontal wells, or wells requiring rigless completions (Wang and Osunjaye, 2016).

PVDF, while mechanically limited, is effective as an inner liner in composite screens, where its acid resistance and low permeability provide protection against scale formation, acid wash degradation, and acid-induced embrittlement. In applications such as water injectors and CO₂ disposal wells, PVDF-lined screens can extend operational life by minimizing chemical attack on metal substrates (Craster et al., 2020).

Inflow Control Devices (ICDs) and Interval Control Valves (ICVs)

ICDs and ICVs demand precise fluid flow regulation, and their internal materials must resist scaling, erosion, and temperature-induced deformation. PEEK has demonstrated effectiveness in high-temperature internal flow control components and ICD restriction elements, where its rigidity and erosion resistance maintain performance under sustained flow rates and thermal cycling (Al Munif et al., 2025).

PPS may serve in chemical barrier layers or internal chambers of ICVs where high-H₂S tolerance and hydrocarbon exposure are expected. Its resistance to oxidative degradation makes it suitable for integration in valves designed for chemical conformance applications, water shutoff, or acid treatment zones (Almuhammadi et al., 2023).

Design and Lifecycle Advantages

Beyond individual tool components, the integration of nonmetallic polymers offers several system-level benefits:

Weight Reduction. Decreased tool mass leads to improved deployment logistics, especially in deviated or ERD wells.

Corrosion Resistance. Reduced reliance on corrosion-resistant alloys (CRAs), lowering overall tool cost and simplifying maintenance.

Thermal and Chemical Customization. Polymers can be tailored by reinforcement or blending to match specific reservoir demands—something far more difficult with metallics.

These benefits suggest a continued trajectory toward hybrid completion systems where nonmetallics are used selectively, optimizing performance without compromising reliability. Future designs are likely to integrate these materials not just as replacements but as core enablers of novel tool architectures.

Conclusion

This comparative analysis evaluated four nonmetallic polymers—PEEK, PVDF, PPS, and SMPs—for their potential integration into HPHT downhole completion tools. Through a review of published performance data, field case studies, and standardized testing protocols, each material was assessed across thermal stability, chemical compatibility, mechanical performance, and deployment behavior.

PEEK consistently demonstrated the most robust all-around profile, withstanding extreme thermal and chemical environments while maintaining high mechanical strength, making it the most suitable for structural and sealing components in packers and high-load completion tools. PPS followed closely, offering excellent chemical resistance and thermal tolerance for barrier applications and hybrid sealing systems. PVDF, while limited in mechanical strength and thermal stability, showed strong performance in acid-prone and sour service environments where it functions effectively as a liner or protective barrier. SMPs, though not structurally load-bearing, introduced a novel deployment paradigm through temperature-triggered expansion, enabling rigless sand control and zonal isolation in complex or extended-reach wells.

Despite these promising attributes, the adoption of nonmetallics in HPHT completions remains constrained by several limitations:

- **Mechanical Load Limits:** PVDF and SMPs lack the compressive and tensile strength needed for high-pressure sealing or structural roles.
- **Long-Term Degradation Data:** Limited availability of extended-duration data under cyclic HPHT and chemically aggressive conditions, especially for SMPs and hybrid composites.
- **Qualification Gaps:** Inconsistent industry standards for nonmetallic material validation in critical wellbore applications create deployment hesitation.
- **Integration Complexity:** Tool design must account for thermal expansion mismatch, bonding interfaces, and pressure cycling behavior, which are more complex in polymer-metal systems.

Future development pathways will likely focus on:

- **Reinforced or hybrid polymer systems** (e.g., PEEK-PTFE or PPS-CF) to improve load-bearing capacity without compromising chemical resistance.
- **Expanded field trials** of SMPs in unconventional or mature wells to validate conformance reliability, filtration longevity, and deployment repeatability.
- **Downhole diagnostics integration**, enabling smart completion tools with embedded SMPs or polymer sensors for adaptive control.
- **Standardization of qualification procedures** (e.g., expanded use of NORSOK M-710 and ISO 23936 frameworks) to streamline polymer deployment in completions.

As the demand for corrosion-resistant, cost-effective, and field-adaptable materials continues to grow, the strategic use of nonmetallic polymers will become increasingly integral to the next generation of intelligent, high-performance completion systems.

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